$$= 1-\{P(A)+P(B)-P(A\cap B)\}$$

$$= 1-P(A)-P(B)+P(A)\cdot P(B)$$

$$= (1-P(A))(1-P(B))$$

$$= P(\overline{A})\cdot P(\overline{B})$$

Thus $\overline{A}$ and $\overline{B}$ are independent .

Similarly one can prove (ii) and (iii). ■

Example 12.18

Two cards are drawn from a pack of forty cards (shown below) in succession.

10 11 12 13 14 15 16 17 18 19	Red in colour
20 21 22 23 24 25 26 27 28 29	Red in colour
10 11 12 13 14 15 16 17 18 19	Black in colour
20 21 22 23 24 25 26 27 28 29	Black in colour

Find the probability that they are 18 and 24 respectively

(i) the first drawn card is replaced (ii) the first drawn card is not replaced

Solution

Let A be the event of drawing a 18 in the first draw.

Let B be the event of drawing a 24 in the second draw.

Case (i)

Card is replaced

$$n(A) = 2, \quad n(B) = 2$$

and $n(S) = 40$ (Total)

Clearly the event A will not affect the probability of the occurrence of event B and therefore A and B are independent.

$$P(A\cap B) = P(A)\,.\,P(B)$$

$$P(A) = \frac{2}{40}\,,\ P(B)=\frac{2}{40}$$

$$P(A\cap B) = P(A)\ P(B)$$

$$= \frac{2}{40}\frac{2}{40}$$

$$= \frac{1}{400}\,.$$

Case (ii)

Card is not replaced

In the first draw, there are two 18 and forty cards in total. Since a 18, drawn at the first draw is not replaced, in the second draw there are only thirty-nine cards in total. Therefore the first event A affects the probability of the occurrence of the second event B. (Note that the event A need not affect the occurrence of event B)

Thus A and B are not independent. That is, they are dependent events.

Therefore, $P(A\cap B) = P(A)\,.\,P(B/A)$

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